Whole grain intake in relation to body weight: from epidemiological evidence to clinical trials.

This viewpoint aims to 1) review the available scientific literature on the relationship between whole grain consumption and body weight regulation; 2) evaluate the potential mechanisms whereby whole grain intake may help reduce overweight and 3) try to understand why epidemiological studies and clinical trials provide diverging results on this topic. All the prospective epidemiological studies demonstrate that a higher intake of whole grains is associated with lower BMI and body weight gain. However, these results do not clarify whether whole grain consumption is simply a marker of a healthier lifestyle or a factor favoring "per se" lower body weight. Habitual whole grain consumption seems to cause lower body weight by multiple mechanisms such as lower energy density of whole grain based products, lower glycemic index, fermentation of non digestible carbohydrates (satiety signals) and finally by modulating intestinal microflora. In contrast with epidemiological evidence, the results of few clinical trials do not confirm that a whole grain low-calorie diet is more effective in reducing body weight than a refined cereal diet, but their results may have been affected by small sample size or short duration of the intervention. Therefore, further intervention studies with adequate methodology are needed to clarify this question. For the time being, whole grain consumption can be recommended as one of the features of the diet that may help control body weight but also because is associated with a lower risk to develop type 2 diabetes, cardiovascular diseases and cancer.